Name	Value	Data Type	Scope
CALC_SP_AbsDelta	0.0	REAL	MainProgram
absolute difference used for comparison logic			
Constant	No		
External Access:	Read/Write		
OPC UA Access:	None		
*CALC_SP_AbsDelta - MainProgram/Main_Lad - *5(ABS), 9(LE)*			
CALC_SP_Delta	0.0	REAL	MainProgram
Target - Ramped			
Constant	No		
External Access:	Read/Write		
OPC UA Access:	None		
*CALC_SP_Delta - MainProgram/Main_Lad - *4(SUB), 5(ABS), 7(GT), 8(LT)*			
CALC_SP_RampRatePersec	1.0	REAL	MainProgram
degrees per second allowed			
Constant	No		
External Access:	Read/Write		
OPC UA Access:	None		
*CALC_SP_RampRatePersec - MainProgram/Main_Lad - *1(MOVE), 3(MUL)*			
CALC_SP_RampStep	0.1	REAL	MainProgram
movement allowed per scan			
Constant	No		
External Access:	Read/Write		
OPC UA Access:	None		
*CALC_SP_RampStep - MainProgram/Main_Lad - *3(MUL), 6(NEG), 7(ADD), 7(GT), 8(LT), 8(SUB), 9(LE)*			
CALC_SP_RampStep_Neg	0	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
OPC UA Access:	None		
*CALC_SP_RampStep_Neg - MainProgram/Main_Lad - *6(NEG)*			
CMD_Auto	1	BOOL	MainProgram
requests Auto mode			
Constant	No		
External Access:	Read/Write		
OPC UA Access:	None		
CMD_Auto - MainProgram/Main_Lad - 12(XIC), 13(XIO)			
CMD_Manual	0	BOOL	MainProgram
requests manual mode			
Constant	No		
External Access:	Read/Write		
OPC UA Access:	None		
CMD_Manual - MainProgram/Main_Lad - 14(XIC), 15(XIO)			
CMD_Start	0	BOOL	MainProgram
enables the loop and process simulation			
Constant	No		
External Access:	Read/Write		
OPC UA Access:	None		
CMD_Start - MainProgram/Main_Lad - 0(XIC)			
CMD_Stop	0	BOOL	MainProgram
disables the loop			
Constant	No		
External Access:	Read/Write		
OPC UA Access:	None		
CMD_Stop - MainProgram/Main_Lad - 0(XIC)			
ONS_AutoReq	0	BOOL	MainProgram

ONS_AutoReq (Continued)
Constant No
External Access: Read/Write
OPC UA Access: None
*ONS_AutoReq - MainProgram/Main_Lad - *12(ONS)*

SIM_AmbientTemp 70.0 REAL MainProgram
room temp baseline
Constant No
External Access: Read/Write
OPC UA Access: None
*SIM_AmbientTemp - MainProgram/Main_Lad - *1(MOVE)*
SIM_AmbientTemp - MainProgram/PlantSim_ST - #7

SIM_HeaterGain 3.0 REAL MainProgram
how strongly CV affects the heater
Constant No
External Access: Read/Write
OPC UA Access: None
*SIM_HeaterGain - MainProgram/Main_Lad - *1(MOVE)*
SIM_HeaterGain - MainProgram/PlantSim_ST - #8

SIM_MaxTemp 300.0 REAL MainProgram
clamp upper bound for the sim
Constant No
External Access: Read/Write
OPC UA Access: None
SIM_MaxTemp - MainProgram/PlantSim_ST - #15, #16

SIM_MinTemp 0.0 REAL MainProgram
clamp lower bound for the sim
Constant No
External Access: Read/Write
OPC UA Access: None
SIM_MinTemp - MainProgram/PlantSim_ST - #19, #20

SIM_TargetTemp 70.0 REAL MainProgram
inteernal temp target calculated from CV
Constant No
External Access: Read/Write
OPC UA Access: None
*SIM_TargetTemp - MainProgram/PlantSim_ST - #12, *#7*

SIM_TaskDT 0.1 REAL MainProgram
task period in seconds
Constant No
External Access: Read/Write
OPC UA Access: None
*SIM_TaskDT - MainProgram/Main_Lad - *1(MOVE), 3(MUL)*
SIM_TaskDT - MainProgram/PlantSim_ST - #12

SIM_TauSeconds 45.0 REAL MainProgram
lag/time constant of the simulated process
Constant No
External Access: Read/Write
OPC UA Access: None
*SIM_TauSeconds - MainProgram/Main_Lad - *1(MOVE)*
SIM_TauSeconds - MainProgram/PlantSim_ST - #12

STS_FirstScanDone 0 BOOL MainProgram
one-time initialization complete
Constant No
External Access: Read/Write
OPC UA Access: None

STS_FirstScanDone (Continued)
*STS_FirstScanDone - MainProgram/Main_Lad - *1(OTL), 1(XIO)*

STS_LoopEnable 0 BOOL MainProgram
master run bit
Constant No
External Access: Read/Write
OPC UA Access: None
*STS_LoopEnable - MainProgram/Main_Lad - *0(OTL), *0(OTU), 19(XIC), 2(XIC)*
STS_LoopEnable - MainProgram/MainRoutine - 0(XIC)

STS_LoopReady 0 BOOL MainProgram
loop is ready to run
Constant No
External Access: Read/Write
OPC UA Access: None
*STS_LoopReady - MainProgram/Main_Lad - *2(OTE), 16(XIC), 17(XIC), 18(XIC), 3(XIC), 4(XIC), 5(XIC), 6(XIC), 7(XIC), 8(XIC), 9(XIC)*

TIC_101_PIDE PID_ENHANCED MainProgram
Constant No
External Access: Read/Write
OPC UA Access: None
*TIC_101_PIDE - MainProgram/PIDE_FBD - *1-A1(IREF,TIC_PV), *1-A1(IREF,TIC_PV), *1-A1(IREF,TIC_PV), *1-A1(IREF,TIC_PV), *1-A1(IREF,TIC_PV), *1-A1(IREF,TIC_PV), *1-A1(IREF,TIC_PV), *1-B1(PIDE,TIC_101_PIDE), *1-D1(OREF,TIC_CV)*
TIC_101_PIDE.SPProg 0.0 REAL
*TIC_101_PIDE.SPProg - MainProgram/Main_Lad - *16(MOVE)*
TIC_101_PIDE.CVProg 0.0 REAL
*TIC_101_PIDE.CVProg - MainProgram/Main_Lad - *17(MOVE)*
TIC_101_PIDE.ProgProgReq 0 BOOL
*TIC_101_PIDE.ProgProgReq - MainProgram/Main_Lad - *10(OTE)*
TIC_101_PIDE.ProgOperReq 0 BOOL
*TIC_101_PIDE.ProgOperReq - MainProgram/Main_Lad - *11(OTU)*
TIC_101_PIDE.ProgAutoReq 0 BOOL
*TIC_101_PIDE.ProgAutoReq - MainProgram/Main_Lad - *12(OTL), *13(OTU)*
TIC_101_PIDE.ProgManualReq 0 BOOL
*TIC_101_PIDE.ProgManualReq - MainProgram/Main_Lad - *14(OTL), *15(OTU)*
TIC_101_PIDE.CV 0.0 REAL
TIC_101_PIDE.CV - MainProgram/Main_Lad - 18(MOVE)

TIC_CV 0.0 REAL MainProgram
actual controller output copied from PIDE
Constant No
External Access: Read/Write
OPC UA Access: None
*TIC_CV - MainProgram/Main_Lad - *18(MOVE)*
*TIC_CV - MainProgram/PIDE_FBD - *1-D1(OREF,TIC_CV), 1-B1(PIDE,TIC_101_PIDE.CVEU)*
TIC_CV - MainProgram/PlantSim_ST - #8

TIC_ManualCV 0.0 REAL MainProgram
manual output command value when in manual mode
Constant No
External Access: Read/Write
OPC UA Access: None
*TIC_ManualCV - MainProgram/Main_Lad - *1(MOVE), 17(MOVE)*

TIC_PV 70.0 REAL MainProgram
simulated temperature process variable
Constant No
External Access: Read/Write
OPC UA Access: None
*TIC_PV - MainProgram/Main_Lad - *1(MOVE)*
TIC_PV - MainProgram/PIDE_FBD - 1-A1(IREF,TIC_PV), 1-B1(PIDE,TIC_101_PIDE.CVProg), 1-B1(PIDE,TIC_101_PIDE.FF), 1-B1(PIDE,TIC_101_PIDE.HandFB), 1-B1(PIDE,TIC_101_PIDE.PV), 1-B1(PIDE,TIC_101_PIDE.RatioProg), 1-B1(PIDE,TIC_101_PIDE.SPCascade), 1-B1(PIDE,TIC_101_PIDE.SPProg)

TIC_PV (Continued)
*TIC_PV - MainProgram/PlantSim_ST - #11, #12, #15, #19, *#11, *#16, *#20*

TIC_SP_Ramped	70.0	REAL	MainProgram

the smoothed setpoint you actually feed into the PIDE

Constant	No
External Access:	Read/Write
OPC UA Access:	None

*TIC_SP_Ramped - MainProgram/Main_Lad - *1(MOVE), *7(ADD), *8(SUB), *9(MOVE), 16(MOVE), 4(SUB), 7(ADD), 8(SUB)*

TIC_SP_Target	150.0	REAL	MainProgram

the operator-entered desired setpoint

Constant	No
External Access:	Read/Write
OPC UA Access:	None

*TIC_SP_Target - MainProgram/Main_Lad - *1(MOVE), 4(SUB), 9(MOVE)*